

Field	Description
Case ID:	CS-2025-001
Case Type:	DLL side-loading via modified file association
Reported by:	Windows Defender
Analyst:	Ali YILMAZ
Date:	2025-10-14
Severity:	Medium
Status:	Closed
Environment:	Production

Case Title

DLL side-loading via modified file association

Case ID

Case-T1546.001_T1574_T1112_T1106

Detection Source

- **Platform:** Microsoft Defender for Endpoint
- **Alert:** *An executable file loaded an unexpected DLL file*
- **Severity:** Medium
- **Category:** Defense Evasion / Persistence

Incident Summary

Defender detected that an executable loaded an unexpected DLL (**libomp140.i386.dll**).
VirusTotal shows 1/72 detection, possible downloader.
Advanced Hunting confirmed the DLL was loaded and file-association keys were modified.

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Tactic	Technique	ID
Persistence	Change Default File Association	T1546.001
Defense Evasion	Modify Registry	T1112
Execution	Native API	T1106
Defense Evasion	Hijack Execution Flow	T1574

Investigation Steps

1. Check **Device name, user, alert time** in Defender portal.
2. Run **Advanced Hunting** queries:
 - DeviceImageLoadEvents → verify DLL loaded.
 - DeviceRegistryEvents → check FileExts / UserChoice changes.
 - DeviceNetworkEvents → see if rundll32.exe or DLL made outbound connections.
3. Review **process tree** (parent + child).
4. Check **file hash, path, signature**.
5. Note if other hosts show same hash.

Evidence

- Defender alert screenshot
- DLL hash (0de51095ffdbfc9ff1632eb2613e6d14ffc5da3f5039dc23cf07262e3297cdc5)
- Hunting results (process + registry events)
- VirusTotal link

Result / Assessment

Check	Status	Notes
DLL Load		Confirmed in ImageLoadEvents
Registry Change		FileExts /UserChoice edited
Network Activity		No outbound found
Verdict	True Positive – Suspicious Persistence Attempt	

Recommended Actions

- Isolate endpoint via Defender.
- Quarantine DLL and related EXE.
- Restore registry (FileExts / UserChoice).
- Run full Defender scan.
- Hunt for same hash on other devices.

Organizational / HR Actions

- Inform IT security team.
- If user ran unverified installer, give awareness training.
- If lab test, close as training.

Business Impact

- Possible unauthorized execution via hijacked file association.
- Limited to one host; medium risk if contained quickly.

Quick Hunting Snippets

Image Load

DeviceImageLoadEvents

where FileName =~ "libomp140.i386.dll"

project Timestamp, DeviceName, InitiatingProcessFileName, FolderPath

Registry Changes

DeviceRegistryEvents

where RegistryKey contains "FileExts" or RegistryKey contains "UserChoice"

project Timestamp, DeviceName, RegistryKey, RegistryValueData

Network Check

DeviceNetworkEvents

where InitiatingProcessFileName has_any ("rundll32.exe", "regsvr32.exe", "libomp")

project Timestamp, DeviceName, RemoteUrl, RemoteIP

The screenshot shows the Microsoft Defender Advanced hunting interface. The left sidebar contains navigation options: Home, Exposure management, Investigation & response (selected), Threat intelligence, Assets, Microsoft Sentinel, Identities, Endpoints, Email & collaboration, Cloud apps, and Cases. The main area is titled "Advanced hunting" and features a "Query" editor. The query is as follows:

```
1 DeviceImageLoadEvents
2 | where FileName startswith "libomp"
3 | summarize count() by FileName, FolderPath, DeviceName
4
```

The results table shows 2 items:

FileName	FolderPath	DeviceName	count
libomp140.i386.dll	C:\Users\Musa\AppData...	desktop-041c16l	1

The screenshot shows the Microsoft Defender Advanced hunting interface. The left sidebar contains navigation options: Home, Exposure management, Investigation & response (selected), Threat intelligence, Assets, Microsoft Sentinel, Identities, Endpoints, Email & collaboration, Cloud apps, and Cases. The main area is titled "Advanced hunting" and features a "Query" editor. The query is as follows:

```
1 DeviceFileEvents
2 | where SHA256 == "0de51895ffdbfc9ff1632eb2613e6d14ffc5da3f5039dc23cf07262e3297cdc5"
3 | distinct DeviceName
4
```

The results table shows 1 item:

DeviceName
desktop-041c16l

Microsoft Defender

Home

Exposure management

Investigation & response

Incidents & alerts

Hunting

Advanced hunting

Custom detection rules

Actions & submissions

Partner catalog

Threat intelligence

Assets

Microsoft Sentinel

Identities

Search

Incidents > Multi-stage incident involving Defense evasion & Discovery on one endpoint

Multi-stage incident involving Defense evasion & Discovery ...

Medium

Active

Unassigned

Go Hunt queries launched from the entity menu now default to a time range starting from the incident's start time up to the execution time. For incidents older than 30 days, the query defaults to last 30-day.

Attack story

Alerts (4)

Assets (2)

Investigations (1)

Evidence and Response (11)

Summary

Alerts

Play attack story

Unpin all

Show all

Oct 14, 2025 9:05 AM

New

Suspicious Security Software Discovery

DESKTOP-04LC16L

Musa

Oct 14, 2025 9:33 AM

Resolved

Automated investigation started manually

desktop-04lc16l

Oct 14, 2025 9:59 AM

New

An executable file loaded an unexpected DLL file

desktop-04lc16l

Musa

Oct 14, 2025 9:59 AM

New

Incident graph

Layout

Group similar nodes

Back to incident details

3 Files

3 Processes

desktop-04lc16l

Microsoft

Communication

Association

3 Processes

Name

svchost.exe -k netswsc -p

"Core-Vec16.exe"

"Ninite.exe" "ea0d6c02631db464661a039e8502c09b44b0b...

 Email notification

Most recent incidents and alerts

Export

Copy link list

Refresh

30 Days

60 Incidents

Search for name or ID

Customize columns

Filter set: Save

Add filter

☐	Incident name	Incident Id	Tags	Severity	Investigation state	Categories	Impacted a
☐	Multi-stage incident involving Defense evasion & ...	270		Medium	2 investigation states	Defense evasion, Discovery...	desktop
☐	An executable file loaded an unexpected DLL file			Medium		Defense evasion	desktop
☐	An executable file loaded an unexpected DLL file			Medium		Defense evasion	desktop
☐	Automated investigation started manually			Informational	No threats found	Suspicious activity	desktops
☐	Suspicious Security Software Discovery			Low		Discovery	DESKTO

Incidents & Alerts

1 active alert, 1 incident



[View Incidents & Alerts](#)

VirusTotal

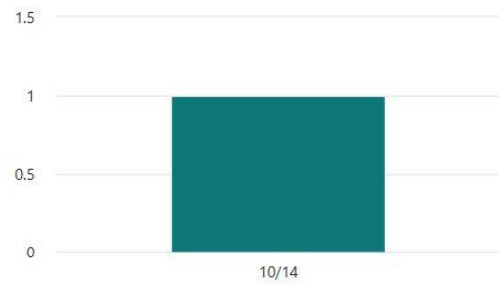
1/72 (High risk)



[View in VirusTotal](#)

Observed devices (last 30 Days)

1 device



Date (First / Last)	Devices
Oct 14, 2025 (First)	desktop-04lc16l

[View all devices](#)

Malware detection



No malware types were found

File names

1 file name

File name	Number of devices
libomp140.i386.dll	1

[View all file names](#)

Observed on cloud apps (last 30 days)



File was not observed on cloud apps

Worldwide observed devices

Low prevalence
1

First seen	Last seen
Oct 14, 2025	Oct 14, 2025

0de51095f9dbcf9f1632eb2613e6d14ffc5da3f5039dc23cf07262e3297cdc5

1

Community Score

-1

1/72 security vendor flagged this file as malicious

Reanalyze Similar More

0de51095f9dbcf9f1632eb2613e6d14ffc5da3f5039dc23cf07262e3297cdc5

libomp140.i386.dll

Size1.26 MB

Last Analysis Date3 days ago

DLL

pedllinvalid-signatureoverlaydetect-debug-environmentsigned

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY1

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis

Do you want to automate checks?

Rising	Downloader.Rugmil1.1337F (CLASSIC)	Acronis (Static ML)	Undetected
AhnLab-V3	Undetected	Alibaba	Undetected
AliCloud	Undetected	ALYac	Undetected
Antiy-AVL	Undetected	Arcabit	Undetected
Arctic Wolf	Undetected	Avast	Undetected

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TECHNIQUES

Change Default File Association

Screensaver

Windows Management

Instrumentation Event Subscription

Unix Shell Configuration Modification

Trap

LC_LOAD_DYLIB Addition

Netsh Helper DLL

Accessibility Features

AppCert DLLs

Adolnit DLLs

Home > Techniques > Enterprise > Event Triggered Execution > Change Default File Association

Event Triggered Execution: Change Default File Association

Other sub-techniques of Event Triggered Execution (17)

Adversaries may establish persistence by executing malicious content triggered by a file type association. When a file is opened, the default program used to open the file (also called the file association or handler) is checked. File association selections are stored in the Windows Registry and can be edited by users, administrators, or programs that have Registry access or by administrators using the built-in assoc utility.^{[1][2][3]} Applications can modify the file association for a given file extension to call an arbitrary program when a file with the given extension is opened.

System file associations are listed under `HKEY_CLASSES_ROOT.[extension]`, for example `HKEY_CLASSES_ROOT.txt`. The entries point to a handler for that extension located at `HKEY_CLASSES_ROOT[handler]`. The various commands are then listed as subkeys underneath the shell key at `HKEY_CLASSES_ROOT[handler]\shell[action]\command`. For example:

- `HKEY_CLASSES_ROOT\txtfile\shell\open\command`
- `HKEY_CLASSES_ROOT\txtfile\shell\print\command`
- `HKEY_CLASSES_ROOT\txtfile\shell\printto\command`

ID: T1546.001

Sub-technique of: T1546

Tactics: Privilege Escalation, Persistence

Platforms: Windows

Contributors: Stefan Kanthak; Travis Smith, Tripwire

Version: 1.1

Created: 24 January 2020

Last Modified: 15 April 2025

Version Permalink

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TECHNIQUES

Hijack Execution Flow

DLL

Dylib Hijacking

Executable Installer

File Permissions Weakness

Dynamic Linker Hijacking

Path Interception by PATH Environment Variable

Path Interception by Search Order Hijacking

Path Interception by Unquoted Path

Home > Techniques > Enterprise > Hijack Execution Flow

Hijack Execution Flow

Sub-techniques (12)

Adversaries may execute their own malicious payloads by hijacking the way operating systems run programs. Hijacking execution flow can be for the purposes of persistence, since this hijacked execution may reoccur over time. Adversaries may also use these mechanisms to elevate privileges or evade defenses, such as application control or other restrictions on execution.

There are many ways an adversary may hijack the flow of execution, including by manipulating how the operating system locates programs to be executed. How the operating system locates libraries to be used by a program can also be intercepted. Locations where the operating system looks for programs/resources, such as file directories and in the case of Windows the Registry, could also be poisoned to include malicious payloads.

ID: T1574

Sub-techniques: T1574.001, T1574.004, T1574.005, T1574.006, T1574.007, T1574.008, T1574.009, T1574.010, T1574.011, T1574.012, T1574.013, T1574.014

Tactics: Persistence, Privilege Escalation, Defense Evasion

Platforms: Linux, Windows, macOS

Version: 1.3

Created: 12 March 2020

Last Modified: 15 April 2025

Version Permalink

Procedure Examples

TECHNIQUES

Modify Registry

- Office Application Startup
- Power Settings
- Pre-OS Boot
- Scheduled Task/Job
- Server Software Component
- Software Extensions
- Traffic Signaling
- Valid Accounts
- Privilege Escalation
- Defense Evasion
- Credential Access
- Discovery

Home > Techniques > Enterprise > Modify Registry

Modify Registry

Adversaries may interact with the Windows Registry as part of a variety of other techniques to aid in defense evasion, persistence, and execution.

Access to specific areas of the Registry depends on account permissions, with some keys requiring administrator-level access. The built-in Windows command-line utility `Reg` may be used for local or remote Registry modification.^[1] Other tools, such as remote access tools, may also contain functionality to interact with the Registry through the Windows API.

The Registry may be modified in order to hide configuration information or malicious payloads via [Obfuscated Files or Information](#).^{[2][3][4][5]} The Registry may also be modified to [Impair Defenses](#), such as by enabling macros for all Microsoft Office products, allowing privilege escalation without alerting the user, increasing the maximum number of allowed outbound requests, and/or modifying systems to store plaintext credentials in memory.^{[6][7]}

The Registry of a remote system may be modified to aid in execution of files as part of lateral movement. It requires the remote Registry service to be running on the target system.^[7] Often [Valid Accounts](#) are required, along with access to the remote system's SMB/Windows Admin Shares for RPC communication.

ID: T1112

Sub-techniques: No sub-techniques

① Tactics: Defense Evasion, Persistence

① Platforms: Windows

Contributors: Bartosz Jerzman; David Lu, Tripwire; Gerardo Santos; Travis Smith, Tripwire

Version: 2.0

Created: 31 May 2017

Last Modified: 15 April 2025

[Version Permalink](#)

TECHNIQUES

Native API

- Scheduled Task/Job
- Serverless Execution
- Shared Modules
- Software Deployment Tools
- System Services
- User Execution
- Windows Management Instrumentation
- Persistence
- Privilege Escalation
- Defense Evasion
- Credential Access
- Discovery

Home > Techniques > Enterprise > Native API

Native API

Adversaries may interact with the native OS application programming interface (API) to execute behaviors. Native APIs provide a controlled means of calling low-level OS services within the kernel, such as those involving hardware/devices, memory, and processes.^{[1][2]} These native APIs are leveraged by the OS during system boot (when other system components are not yet initialized) as well as carrying out tasks and requests during routine operations.

Adversaries may abuse these OS API functions as a means of executing behaviors. Similar to [Command and Scripting Interpreter](#), the native API and its hierarchy of interfaces provide mechanisms to interact with and utilize various components of a victimized system.

Native API functions (such as `NtCreateProcess`) may be directed invoked via system calls / syscalls, but these features are also often exposed to user-mode applications via interfaces and libraries.^{[3][4][5]} For example, functions such as the Windows API `CreateProcess()` or GNU `fork()` will allow programs and scripts to start other processes.^{[6][7]} This may allow API callers to execute a binary, run a CLI command, load modules, etc. as thousands of similar API functions exist for various system operations.^{[8][9][10]}

ID: T1106

Sub-techniques: No sub-techniques

① Tactic: Execution

① Platforms: Linux, Windows, macOS

Contributors: Gordon Long, Box, Inc., @ethicalhax; Stefan Kanthak; Tristan Madani (Cybereason)

Version: 2.3

Created: 31 May 2017

Last Modified: 15 April 2025

[Version Permalink](#)